HYPERBOLIC INVERSE REINFORCEMENT LEARNING

Iulia Rusu
University of California San Diego
Salk Institute for Biological Studies
irusu@ucsd.edu

Anoop Praturu
Causal Labs
anoop@getcausal.ai

Sergei Gepshtein
University of California San Diego
sgepshtein@ucsd.edu

Tatyana O. Sharpee
University of California San Diego
Salk Institute for Biological Studies
sharpee@salk.edu

arXiv:submit/8285942 [cs.LG] 30 Oct 2026

ABSTRACT

Inverse reinforcement learning (IRL) seeks to infer the reward function underlying observed behavior, but its predictive accuracy and interpretability depend on how that reward is computed from the underlying features. While prior IRL reward functions have been implemented through a variety of techniques including linear weights, neural networks, or other function approximators, none make assumptions about the geometry of the underlying feature space. Inspired by geometric organization observed in neural circuits, we introduce Hyperbolic inverse reinforcement learning (Hyperbolic IRL), which parameterizes reward as a function of Hyperbolic distance between the feature vector and a learned parameter vector, as well as its Euclidean distance variant. We tested our methods against a range of linear and nonlinear IRL baselines under both deterministic and stochastic policies. Our Hyperbolic approach achieved comparable performance to state-of-the-art models on established benchmarks and excelled in the low data limit and across dense environments. Applied to human navigation among multiple exhibits in a museum room, Hyperbolic IRL reveals reward structure that depends not only on individual exhibit-distance features, but also on non-separable conjunctions between them. Together, these results demonstrate that a Hyperbolic reward parameterization can improve both the predictive accuracy and interpretability of IRL, providing a framework for studying how non-separable, conjunctive structure can begin to explain behavior.

1 INTRODUCTION

Inverse Reinforcement Learning (IRL) methods learn a reward function from expert demonstrations, formulated within a Markov Decision Process (MDP) where states are represented through a feature function and reward is recovered over these features rather than the states themselves. Learning a generalizable policy this way is challenging: many reward functions can explain the demonstrations equally well, yet correspond to policies that overfit and fail to generalize (Ng & Russell, 2000). Thus, it is important to not only explain the demonstrations, but to do so with an interpretable reward structure. Existing methods impose such structure by fixing how reward is computed from features: classic approaches take reward to be a linear combination of hand-selected features (Ng & Russell, 2000; Abbeel & Ng, 2004), while more recent work replaces this linear combination with a more expressive function approximator such as a Gaussian process (Levine et al., 2011) or a neural network (Wulfmeier et al., 2015). Whether linear or nonlinear, however, these approaches all assume the underlying feature space is flat.

As environments grow more complex, so does the dimension of the feature and reward space. A common assumption in ML is that despite this high dimension, correlations among features let the data be described by a lower-dimensional manifold. But the right geometry for that compression, not just its dimension, depends on the structure of the data (Nickel & Kiela, 2017). Assuming a flat manifold when the true structure is curved can distort the compression. In particular, hyperbolic space

is known to pack substantially more structure into a small number of dimensions than a flat space of the same dimension can: because its volume grows exponentially rather than polynomially with radius, it can embed hierarchical or exponentially-branching structure at far lower distortion using far fewer dimensions (Cannon et al., 1997; Krioukov et al., 2010; Nickel & Kiela, 2017). Extending standard machine learning methods to curved manifolds, however, is not straightforward. Most methods, such as matrix factorization (Mnih & Salakhutdinov, 2007) and neural networks (Vaswani et al., 2017), measure similarity between vectors using the dot product, which is a fundamentally Euclidean notion with no natural analogue on a curved space (Cannon et al., 1997). Distance, by contrast, is well defined on any Riemannian manifold, flat or curved. Praturu & Sharpee (2025) show that replacing the dot product with a distance-based similarity measure not only makes this extension to curved geometries possible, but is itself a more expressive and better-generalizing choice than the dot product even when curvature is held fixed, a pattern also found in other domains, e.g. neural population encoding (Wu et al., 2026) and drug-target interaction prediction (Poleksic, 2023). We adopt the same distance-based approach for reward in IRL: reward is computed as distance between the feature vector and a learned parameter vector, so that the same functional form can be instantiated in either a Euclidean or a Hyperbolic feature space without any change to its basic structure. Having multiple curvature options allows one to better fit to the true shape of the reward manifold.

Our distance-based geometric approach was also inspired by the Hyperbolic organization found in neural circuits (Zhang et al., 2023; Zhou et al., 2018). Meanwhile IRL itself is an increasingly common tool for analyzing behavior and decision-making in neuroscience (Ashwood et al., 2022; Ke et al., 2025), so a reward geometry grounded in the brain's own representational structure is a natural fit for this growing use case. Despite this, Hyperbolic distance has not previously been used for IRL, although Hyperbolic geometry has been applied broadly elsewhere in ML: to hierarchical embeddings (Nickel & Kiela, 2017), neural network architectures (Ganea et al., 2018), graph representation learning (Chamberlain et al., 2017), and the geometry of complex networks (Krioukov et al., 2010).

Our contribution is a framework for Distance-Based Inverse Reinforcement Learning, based on Maximum Entropy IRL (Ziebart et al., 2008). We define the reward to be a function of distance between the feature vector and learned parameter vector in either Hyperbolic or Euclidean space. We argue that the distance-based reward, and the Hyperbolic variant in particular, offers several advantages: the flexibility to select a curvature suited to the true geometry of the reward, a substantial reduction in the number of parameters needed to achieve competitive performance relative to more heavily parameterized function approximators, a closer biological relevance to the geometry observed in biological neural circuits, and an interpretable reward structure as a function of joint and separate feature components.

2 Method: Distance-based Rewards

2.1 MDPs and Maximum Entropy IRL

We consider a Markov Decision Process $\mathcal{M} = \{\mathcal{S}, \mathcal{A}, \mathcal{T}, \gamma\}$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $\mathcal{T} : \mathcal{S}\times\mathcal{A}\times\mathcal{S} \to [0, 1]$ is the transition distribution, and $\gamma \in [0, 1]$ is the discount factor. Inverse reinforcement learning (IRL) takes in observed expert demonstrations $\mathcal{D} = \{\tau_1, \ldots, \tau_N\}$, with each consisting of state and action pairs $\tau_i = \{(s_1, a_1), (s_2, a_2), \ldots\}$. There is a reward value associated with each state $R(s_t)$. IRL tries to find the reward function R responsible for the set of expert demonstrations.

While there are many IRL frameworks, including margin-based (Ratliff et al., 2006), Bayesian (Ramachandran & Amir, 2007), and adversarial (Fu et al., 2018), we use Maximum Entropy IRL (MaxEnt IRL) (Ziebart et al., 2008), which handles expert suboptimality and environmental stochasticity by modeling a full distribution over trajectories rather than assuming the expert acts optimally. In MaxEnt IRL, the state s_t is described by a feature vector $\vec{\phi}(s_t)$. The framework states that among all trajectory distributions whose expected features match the expert's observed feature expectations, the one with maximum entropy is the most correct. This yields a distribution over trajectories that is exponential in the accumulated reward:

$$P(\tau \mid \theta) = \frac{1}{Z(\theta)} \exp\left(R(\tau)\right), \qquad R(\tau) = \sum_{t=1}^{T} R(s_t), \tag{1}$$

where $Z(\theta) = \sum_\tau \exp(R(\tau))$ is the partition function. θ are the parameters of the reward function. The reward function for a single state was originally written as a function of the state's features $\vec{\phi}(s_t)$ and the learned $\vec{\theta}$ vector, with the assumption that reward is linear in these features (Ziebart et al., 2008).

$$R(s_t) = \vec{\phi}(s_t) \cdot \vec{\theta} \tag{2}$$

Fitting for θ by maximum likelihood gives the general IRL loss

$$\mathcal{L}(\theta) = -\frac{1}{N} \log P(\mathcal{D} \mid \theta) = \log Z(\theta) - \frac{1}{N} \sum_{i=1}^{N} R(\tau_i), \tag{3}$$

which is convex in θ whenever $R(\tau)$ is linear in θ, and whose gradient is:

$$\nabla_\theta \mathcal{L}(\theta) = \mu_\pi - \mu_E, \tag{4}$$

where $\mu_\pi = \mathbb{E}_{\tau \sim \pi_\theta} [\nabla_\theta R(\tau)]$ denotes the expected reward gradient under the trajectory-visitation distribution of the policy, and $\mu_E = \frac{1}{N} \sum_{i=1}^{N} \nabla_\theta R(\tau_i)$ denotes the corresponding empirical average over the expert demonstrations.

What varies across our proposed models is the parametric form of R in Equation 2; we introduce the notion of "distance" between a state's features $\vec{\phi}(s_t)$ and the reward parameters $\vec{\theta}$.

2.2 Euclidean Distance

We take $R(s_t) = -\frac{1}{2} d^2$, the negative squared Euclidean distance between $\vec{\phi}(s_t)$ and $\vec{\theta}$, which expands to a linear-in-features form

$$R(s_t) = \vec{\phi}(s_t) \cdot \vec{\theta} - \tfrac{1}{2} \left(\|\vec{\phi}(s_t)\|^2 + \|\vec{\theta}\|^2 \right). \tag{5}$$

Substituting into Equation 3 and dropping the constant $\|\vec{\phi}\|^2$ term gives

$$\mathcal{L}(\theta) = \log Z(\theta) - \mathbb{E}_{\mu_E} \left[\vec{\phi}(s_t) \cdot \vec{\theta} - \tfrac{1}{2} \|\vec{\theta}\|^2 \right], \tag{6}$$

with gradient

$$\nabla_\theta \mathcal{L}(\theta) = \mathbb{E}_{\mu_\pi} \left[\vec{\phi}(s_t) - \vec{\theta} \right] - \mathbb{E}_{\mu_E} \left[\vec{\phi}(s_t) - \vec{\theta} \right]. \tag{7}$$

The full step-by-step derivation is given in Appendix A.1.

2.3 Hyperbolic Distance

We keep $R(s_t) = -\frac{1}{2} d^2$, but now measure d in Hyperbolic space. We use the Lorentz model of Hyperbolic space $\mathbb{H}^d$: the upper sheet of a two-sheeted hyperboloid embedded in $(d+1)$-dimensional Minkowski space, equipped with the Minkowski inner product $\langle \vec{u}, \vec{v} \rangle_{\mathcal{M}} = -u_0 v_0 + \sum_{n=1}^{d} u_n v_n$. A point $\vec{x} \in \mathbb{R}^{d+1}$ lies on $\mathbb{H}^d$ exactly when $\langle \vec{x}, \vec{x} \rangle_{\mathcal{M}} = -1$ and $x_0 > 0$.

Our feature vector $\vec{\phi}(s_t)$ and the learned parameter vector $\vec{\theta}$ are both d-dimensional, one dimension short of the ambient space $\mathbb{R}^{d+1}$ that $\mathbb{H}^d$ lives in. To put them into Hyperbolic space we add an additional component to each. Given a d-dimensional vector $\vec{v}$, we set $v_0 = \sqrt{1 + \|\vec{v}\|^2}$ and take $(v_0, \vec{v}) \in \mathbb{R}^{d+1}$ as the corresponding point on $\mathbb{H}^d$, since $\langle (v_0, \vec{v}), (v_0, \vec{v}) \rangle_{\mathcal{M}} = -v_0^2 + \|\vec{v}\|^2 = -1$ by construction. Applying this to both $\vec{\phi}(s_t)$ and $\vec{\theta}$ gives $\phi_0(s_t) = \sqrt{1 + \|\vec{\phi}(s_t)\|^2}$ and $\theta_0 = \sqrt{1 + \|\vec{\theta}\|^2}$, so that θ now sits on the hyperboloid in this $(d+1)$-dimensional space, alongside $\vec{\phi}(s_t)$.

The distance between the two points on the hyperboloid is

$$d = \operatorname{arcosh} \left(-\langle \vec{\theta}, \vec{\phi}(s_t) \rangle_{\mathcal{M}} \right), \tag{8}$$

so that

$$R(s_t) = -\tfrac{1}{2} \left[\operatorname{arcosh} \left(-\langle \vec{\theta}, \vec{\phi}(s_t) \rangle_{\mathcal{M}} \right) \right]^2. \tag{9}$$

This can now be optimized directly in hyperbolic space using Riemannian gradient descent, the details of which we leave to Appendix A.2.

2.4 Generalizing to Arbitrary Curvature

The formulation above fixes the hyperboloid to unit radius, $\langle \vec{\theta}, \vec{\theta} \rangle_{\mathcal{M}} = -1$, corresponding to constant curvature -1. We generalize to a hyperboloid of radius R (constant curvature $-1/R^2$) by rescaling the defining constraint to $\langle \vec{\theta}, \vec{\theta} \rangle_{\mathcal{M}} = -R^2$, i.e. $\theta_0 = \sqrt{R^2 + \|\vec{\theta}_{1:d}\|^2}$, so that the distance between two points becomes

$$d_R\left(\vec{\theta}, \vec{\phi}(s_t)\right) = R \operatorname{arcosh}\left(-\frac{\langle \vec{\theta}, \vec{\phi}(s_t) \rangle_{\mathcal{M}}}{R^2}\right), \tag{10}$$

with reward $R(s_t) = -\frac{1}{2} d_R^2$ as before. We denote this constant curvature $\kappa = -1/R^2$, and refer to a given radius-R model as Hyperbolic ($\kappa = -1/R^2$) throughout the rest of the paper. Riemannian gradient descent generalizes in a similar way, with the gradient correction and exponential map each picking up a factor of R; full details in Appendix A.3. This lets us treat curvature as a hyperparameter and compare reward geometries across a range of curvatures rather than fixing it to -1.

3 Results

We tested the performance of our distance-based reward models using the Objectworld benchmark (Levine et al., 2011), as well as a real world benchmark.

3.1 Objectworld: Ground Truth Reward Recovery

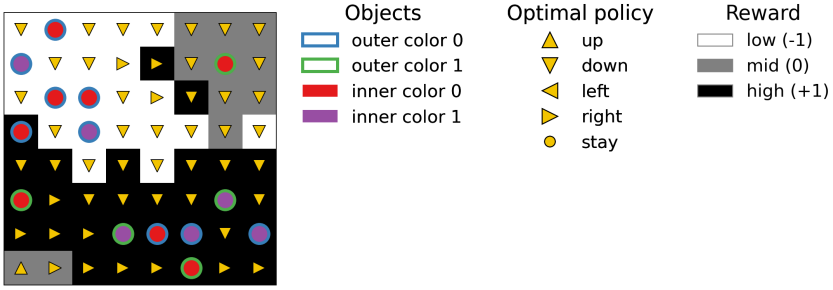


Figure 1: Example 8×8 Objectworld environment with 14 objects, $C = 2$ colors. True reward is $+1$ (black) within 3 cells of an outer-color-0 object AND within 2 cells of an outer-color-1 object, -1 (white) within 3 cells of an outer-color-0 object alone, and 0 (mid gray) otherwise. Yellow arrows show the optimal policy under the reward.

Objectworld is an $N \times N$ grid on which n_{objects} objects are placed on random, non-overlapping cells. Each object is independently assigned an inner and an outer color drawn uniformly from C colors; every color contributes a feature component to the state description, but only two of the C outer colors are ever relevant to determining the reward, with every other color, including all inner colors, acting as a distractor feature that plays no role in the reward function.

Each grid cell is represented by a $2C$-dimensional feature vector: the Euclidean distance from the cell to the nearest object of each of the C outer colors, followed by the distance to the nearest object of each of the C inner colors (a color with no placed object gets a constant distance larger than any true in-grid distance, instead of ∞). The ground truth reward is a nonlinear function of only two of these $2C$ features, the distance to the nearest outer-color-0 and outer-color-1 object: $+1$ near both, -1 near outer-color-0 alone, 0 otherwise. There is built-in stochasticity as selection of action has a 30% chance of being selected randomly, the other 70% of time, the action is the intended action. We illustrate a small version of the grid in Figure 1.

We tested our 3 distance-based reward functions: Euclidean, Hyperbolic ($\kappa = -1$) and Hyperbolic ($\kappa = -4$). We compare them to classical Maximum Entropy Inverse Reinforcement Learning (MaxEnt IRL) (Ziebart et al., 2008) and DeepIRL (Wulfmeier et al., 2015). We compare the

performance of these models using the Expected Value Difference (EVD) metric, which measures how suboptimal the policy is under the true reward. To do this, we use the learned reward from each respective method and compute the optimal deterministic policy using $\arg\max_a Q(s, a)$ under that reward. We then measure the expected sum of discounted rewards under the policy. We also evaluate under a stochastic policy, induced by soft value iteration on each learned reward, $\pi(a \mid s) \propto \exp(Q_{\text{soft}}(s, a))$.

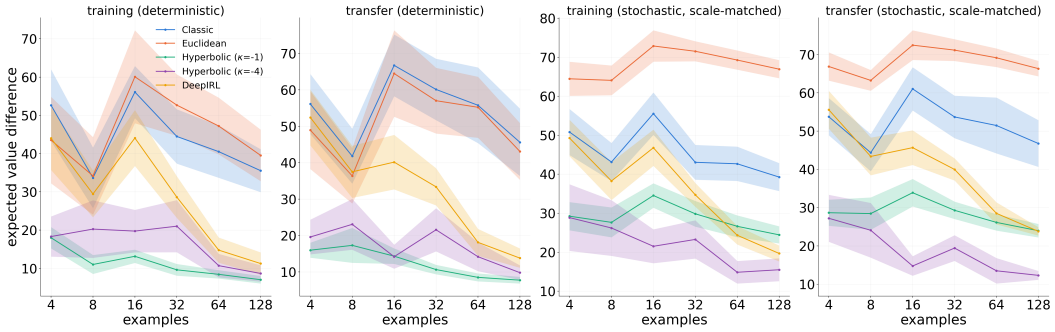


Figure 2: Expected value difference vs. number of demonstration examples on 32×32 Objectworld with 50 objects, Classic, Euclidean, Hyperbolic ($\kappa = -1$), Hyperbolic ($\kappa = -4$), and a neural network reward (16 repeats per condition). Hyperbolic Distance with negative curvature of $\kappa = -4$ had the best performance with increased sample count.

We report the EVD comparison as the number of examples from the ground truth policy used for training increases, in Figure 2. We also evaluate the policies on a transfer to 4 held-out environments, where we change the location of objects in the grid, but not the size or the number of objects. We observe that both Hyperbolic models outperform the others, with Hyperbolic ($\kappa = -4$) showing the best performance with increased examples in training, across both stochastic and deterministic policies and in transfer environments. In Figure 3, we see that both Hyperbolic variants lead in performance, with increasing object density in the grid.

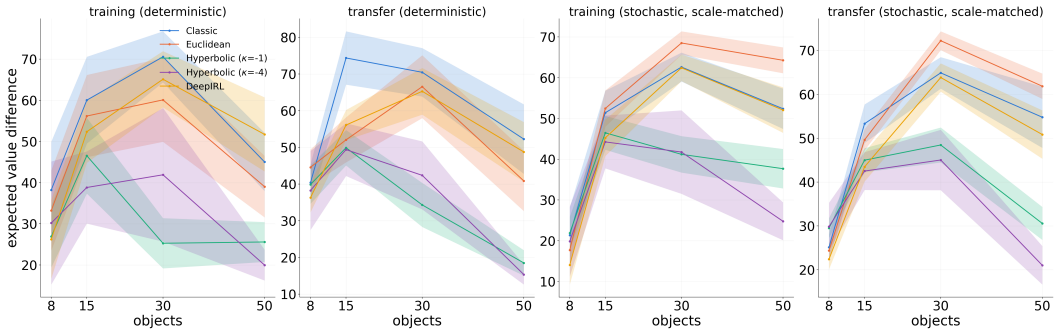


Figure 3: Expected value difference vs. number of objects (8, 15, 30, and 50 objects, corresponding to approximately 0.8%, 1.5%, 2.9%, and 4.9% grid-cell density) on 32×32 Objectworld, Classic, Euclidean, Hyperbolic ($\kappa = -1$), Hyperbolic ($\kappa = -4$), and DeepIRL. Hyperbolic Distance with curvature of $\kappa = -4$ showed the best performance out of all, with increased object density.

3.2 MUSEUM

We test model performance on human behavioral data from HUMEX, a dataset of visitor movement in exhibition spaces. The data used here were collected in a ceramics gallery at the Los Angeles County Museum of Art (LACMA), where ten unobtrusive cameras mounted along the upper perimeter walls recorded visitor movement during museum opening hours over several months. The recordings were processed to recover continuous two-dimensional visitor trajectories in a common ground-plane coordinate system. The resulting trajectory data are publicly available through the

HUMEX repository on OSF (Gepshtein et al., 2026) (`https://osf.io/7f5du/`). Figure 4A shows example trajectories from the dataset. We discretized the floorplan into grid cells of $0.3m$ per side. The action space consisted of five actions: stay, up, down, left, and right. In the feature vector, the first 13 dimensions were the normalized Euclidean distance from the agent to each respective exhibit; the last two were the agent's own normalized x and y position in the space. We trained on 18 of a total of 26 recorded trajectories, holding out 4 for validation and 4 for testing.

We quantified policy recovery using Average Displacement Error (ADE) and Final Displacement Error (FDE) which are standard metrics from the human trajectory-forecasting literature (Alahi et al., 2016; Xu et al., 2018). ADE measures, for each true trajectory, the mean Euclidean distance to 10 independent rollouts of the learned policy, each starting from the trajectory's own initial state. We averaged ADE from each true trajectory to get a final score. For FDE we do the same, but only use the final time point. The FDE is important in this specific scenario as all the trajectories in the analysis return to their initial location at the exit of the museum. Together these two measurements test how well the methods recapitulate global patterns in the data.

In Figure 4B-C we note that Hyperbolic $\kappa = -4$ had the best ADE and FDE performance out of all the models. Notably, it achieves this with 15 reward-function parameters, compared to 1,601 for DeepIRL.

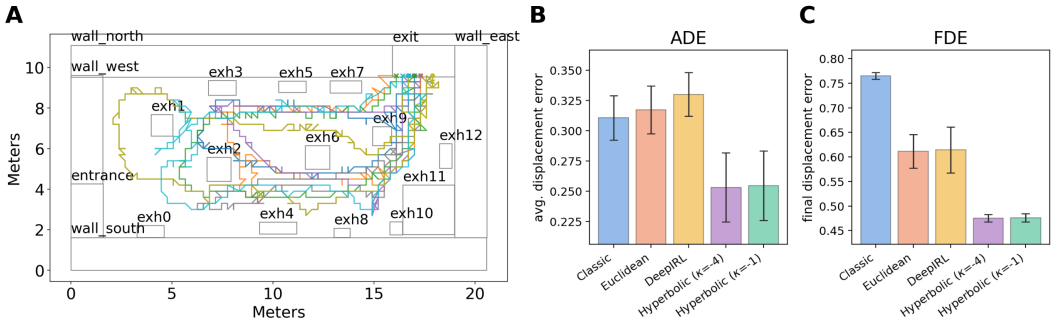


Figure 4: A: Floorplan of LACMA ceramics gallery's 13 exhibits, with example walking trajectories colored by individual. B, C: Average Displacement Error and Final Displacement Error of models trained on museum trajectories, error bars showing standard error of the mean across the 4 held-out test trajectories. Hyperbolic ($\kappa = -4$) is best at modeling the average trajectory and its starting and final locations.

3.3 Feature Coupling

To test whether each reward model captures genuine interactions between feature components, rather than treating every feature component as an independent additive contribution to reward, we probed the fitted reward function directly. For a pair of exhibit-distance feature components, we hold every other feature at its median value and sweep ϕ_i, ϕ_j over $[0, 1]$, plotting the resulting reward surface $R(\theta, \phi)$ in Figure 5.

We measured interaction using a two-way ANOVA-style fit: $R(\phi_i, \phi_j) \approx c + \text{row-effect}(\phi_i) + \text{col-effect}(\phi_j)$, where the row/column effects are the deviations of each row/column mean from the overall mean c. This additive approximation is what a reward model would produce if ϕ_i and ϕ_j contributed to the reward independently of one another; the residual, $R - (c + \text{row-effect} + \text{col-effect})$ (Figure 5, bottom row), is the part of the reward surface that no additive combination of main effects can explain. A near-zero residual indicates the two features act independently under that reward model, while visible structure in the residual indicates a coupling between them. We saw that the Hyperbolic models were the only ones exhibiting this non-flat residual, and therefore were experiencing non-separable feature coupling.

To summarize this non-additivity as a single number per feature pair, we also compute the mixed second difference of the reward surface (full definition in Appendix A.5). For ease of reference, and because of its similarity to the concept of conjunctive coding in neuroscience, we will refer to this quantity throughout the rest of the paper as the *conjunctive effect*.

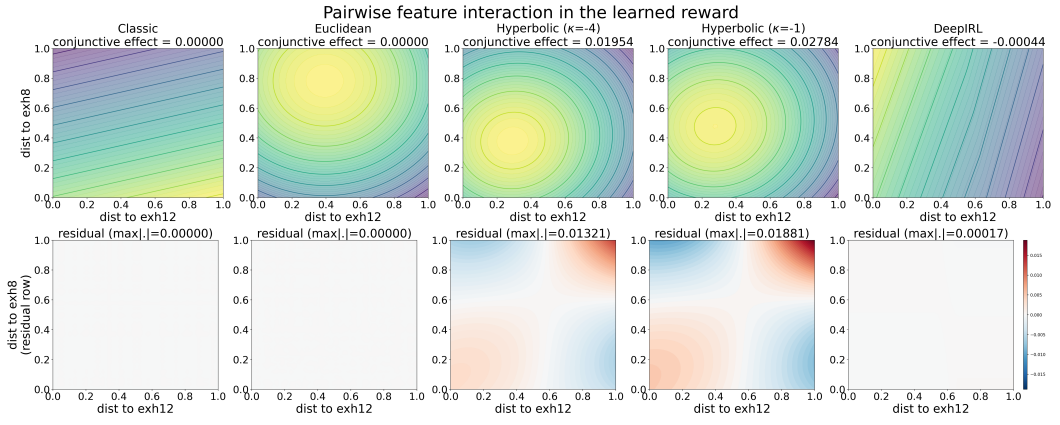


Figure 5: Reward as a function of two feature dimensions (distance to exhibit 12 and distance to exhibit 8), with the remaining 13 feature dimensions held at their median values, for each fitted reward model. Top row: raw reward surface, as a function of two features. Bottom row: residual after removing the best additive (two-way ANOVA main-effects) approximation of the surface.

In Figure 6 we analyze the conjunctive effect of the Hyperbolic ($\kappa = -4$) model, the geometry with the best FDE and ADE performance on the museum dataset. We repeated the same conjunctive-effect probe used above for a single pair of exhibits, for every pair of exhibit-distance feature components, shown as the matrix in Figure 4A. Probing for the strongest conjunctive effects, we saw clusters around exhibit 12 and exhibit 3. Meanwhile the weakest all involved exhibit 6.

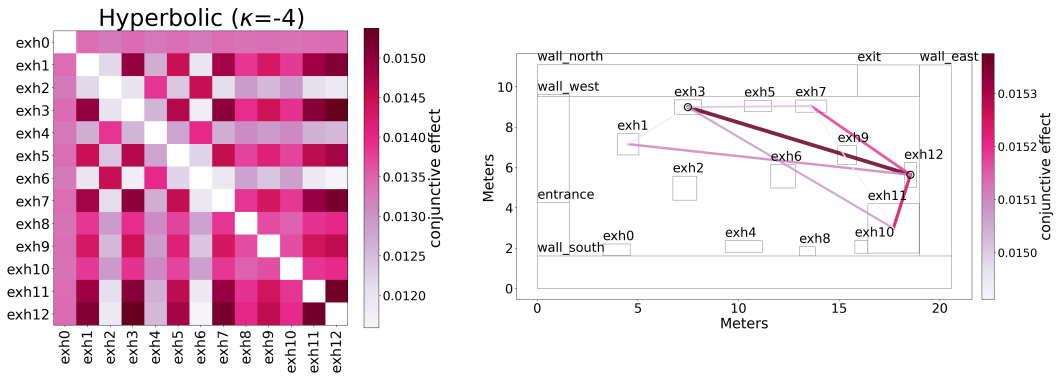


Figure 6: Pairwise exhibit-distance interaction for the Hyperbolic ($\kappa = -4$) model. Left: the same conjunctive effect measure as Figure 5, computed for all exhibit pairs. Right: the 8 strongest conjunctive effects drawn on the museum floorplan, line width and color scaled to the conjunctive effect; exhibit 3 and exhibit 12 (circled) are hubs that every one of the 8 strongest pairs routes through.

To test whether this fitted coupling of the conjunctive effect reflects genuine visitor behavior rather than an artifact of the reward parameterization, we construct a purely behavioral measure, $\mathrm{co_attendance}(i, j)$, that does not use the fitted reward at all. For each visitor, we find how close they came to each exhibit at any point during their visit; for a given pair of exhibits i, j, $\mathrm{co_attendance}(i, j)$ is then the correlation, across all visitors, between how close they got to exhibit i and how close they got to exhibit j. A positive value indicates that visitors who sought out one exhibit tended to also seek out the other. Figure 7 plots, for each geometry, the fitted conjunctive effect for all 78 exhibit pairs against the real $\mathrm{co_attendance}$ value for that pair, with a fitted line, Pearson r, and permutation-corrected p-value per panel. Because the 78 pairs are built from only 13 exhibits and are therefore not independent samples, we assess significance with a permutation test rather than the naive p-value. The full methodology, naive-vs-corrected comparison, and a distance-controlled partial-correlation check are reported in Table 1 (Appendix A.7). All three hyperbolic

curvatures show a significant positive relationship; Classic and Euclidean show essentially none, and this pattern is unchanged after controlling for physical exhibit-to-exhibit distance, ruling out mere proximity as an explanation (Appendix A.7).

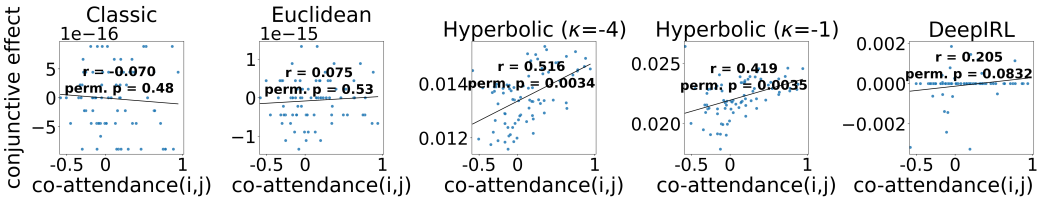


Figure 7: Fitted feature interaction vs. real exhibit co-attendance, per geometry, for all 78 exhibit pairs (allobjects), with a fitted line, Pearson r, and permutation-corrected p-value (10,000 shuffles of exhibit labels) shown per panel. Classic and Euclidean show no relationship; both Hyperbolic variants shown here have a significant positive relationship (full naive-vs-corrected comparison and a distance-controlled check in Table 1, Appendix A.7).

4 Discussion and Future Work

In this work we introduce Hyperbolic IRL, a novel reward parameterization that improves both predictive accuracy and interpretability while maintaining a low number of parameters, excelling especially in low-data and high-density settings. Applied to museum visitor trajectories, we found that the inferred reward contains a conjunctive effect between exhibit-distance features that corresponds to co-attendance in the observed behavior, connecting to the neuroscience literature on conjunctive coding.

We offer two possible reasons for the Hyperbolic model's advantage over other models. Firstly, Hyperbolic space may be intrinsically better suited to representing densely distributed reward-relevant objects. It is a classical fact that the volume of a ball in Hyperbolic space grows exponentially with its radius, rather than polynomially as in Euclidean space (Cannon et al., 1997), a property already exploited to justify Hyperbolic geometry for hierarchical and scale-free structures (Krioukov et al., 2010; Nickel & Kiela, 2017). Praturu & Sharpee (2025) note that this lets points be packed densely around a reference location while remaining well-separated under the metric the model actually uses. In our setting, this means each object's zone of high reward can stay sharply localized around it without overlapping neighboring objects' zones, even as those objects become more numerous. This offers an explanation for its advantage specifically as object density increases (Figure 3). Secondly, the Hyperbolic reward's coupling between feature components offers a complementary explanation. Section 3.3 showed that the Hyperbolic distance method captures a genuine conjunctive effect between exhibit-distance features, and that this conjunctive effect tracks real visitor behavior, unlike the additive Classic, Euclidean, and DeepIRL rewards. The reward's joint dependence on multiple features at once is unique to our model, but not a unique concept.

Indeed, selectivity for combinations of stimulus features, rather than single features in isolation, is a well-established property of sensory coding: classic characterizations of complex cells in visual cortex and more general spike-triggered analyses both show that sensory neurons are typically best described by several jointly-relevant stimulus features nonlinearly combined, rather than by one feature alone (Adelson & Bergen, 1985; Rust et al., 2005; Schwartz et al., 2006; Bialek & de Ruyter van Steveninck, 2005; Sharpee, 2013). The conjunctive structure we find in the museum reward function is consistent with this broader pattern of feature-combination sensitivity in biological sensory systems. It may prove to be an advantageous tool for behavioral neuroscience research.

There has been a body of neuroscience research that focuses on studying decision-making and behavior using IRL (Ashwood et al., 2022; Ke et al., 2025; Schultheis et al., 2021; Yamaguchi et al., 2018; Kwon et al., 2020; Reddy, 2022). However, much of the field remains bounded to studying behavior in artificial, simplified settings, due to the relative ease of applying analytical tools in these more tractable contexts (Ashwood et al., 2022; Carandini & Churchland, 2013). We argue that the Hyperbolic method's unique advantages in performance in dense-object settings with multiple reward sources and an underlying nonlinear reward structure make it a good candidate over other,

more high-parameter methods such as neural-network-based models. We reported each model's parameter count in the Results section: 1,601 for DeepIRL versus 15 for every other model. A formal model-complexity comparison (e.g., BIC) would require $N \gg K$ data points relative to parameters, which does not hold here given DeepIRL's parameter count relative to our training set size, so we do not report one. Nonetheless, the Hyperbolic model's substantially lower parameter count and its capacity to perform well in the low-data limit, as demonstrated in Figure 2, make it a good candidate for behavioral research, which is often constrained by limited sample sizes (Roy et al., 2021; Wilson & Collins, 2019). A concrete example of a good use case for this method is foraging: the study of how animals search for and exploit resources, weighing decisions such as when to leave a depleting patch for a potentially better one elsewhere (Stephens & Krebs, 1986; Mobbs et al., 2018; Dal Bello et al., 2021; Calhoun et al., 2015). Arguably, the museum dataset we presented is itself a type of foraging task, with visitors searching among and choosing between multiple exhibits (reward sources) distributed through space.

In future work, we will modify our model to learn the curvature of the Hyperbolic space directly via a Bayesian approach, as was done for hyperbolic data embedding by Praturu & Sharpee (2024), rather than testing over discrete sets of values. Another important direction will be testing whether the conjunctive selectivity observed here generalizes to other real-world behavioral datasets, and, where paired neural recordings are available, whether it aligns with conjunctive coding in recorded neural activity.

AI USE STATEMENT

In this work, we used generative AI tools for refining our literature search, grammar help, and writing plotting code to generate scientific figures. We have reviewed all AI-assisted work. We checked LLM-generated research ideas for potential plagiarism through a manual literature survey and verified that the plotting of figures was done correctly. We have not used generative AI tools to generate synthetic data sets, help develop theoretical models or conceptual frameworks, formulate mathematical claims, provide critical ingredients for proving mathematical claims, propose or refine hypotheses, design or provide feedback on research methodology or experiments, implement methods, clean and reformat datasets, support qualitative and thematic data analysis, and interpret results. The rest of the required disclosure tasks such as assistance in the writing of proofs, and assistance with translation are not applicable to this work. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

ETHICS STATEMENT

This work uses human visitor trajectory data from HUMEX, a publicly available dataset (Gepshtein et al., 2026) (`https://osf.io/7f5du/`) collected via unobtrusive overhead cameras in a public gallery at the Los Angeles County Museum of Art. We only had access to de-identified, two-dimensional visitor trajectories; no raw video, individual identities, or other personally identifiable information was available to us. The Salk Institute Institutional Review Board granted exemption approval for this project on January 7, 2022, under Common Rule 45 CFR 46.104(d)(2). We do not foresee other ethical concerns regarding methodology, dual use, or fairness arising from this work.

REFERENCES

Pieter Abbeel and Andrew Y. Ng. Apprenticeship learning via inverse reinforcement learning. In *Proceedings of the Twenty-First International Conference on Machine Learning (ICML)*, pp. 1–8, 2004.

Edward H. Adelson and James R. Bergen. Spatiotemporal energy models for the perception of motion. *Journal of the Optical Society of America A*, 2(2):284–299, 1985.

Alexandre Alahi, Kratarth Goel, Vignesh Ramanathan, Alexandre Robicquet, Li Fei-Fei, and Silvio Savarese. Social LSTM: Human trajectory prediction in crowded spaces. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 961–971, 2016.

Zoe Ashwood, Aditi Jha, and Jonathan W. Pillow. Dynamic inverse reinforcement learning for characterizing animal behavior. In *Advances in Neural Information Processing Systems 35 (NeurIPS)*, 2022.

William Bialek and Rob R. de Ruyter van Steveninck. Features and dimensions: Motion estimation in fly vision. *arXiv preprint arXiv:q-bio/0505003*, 2005.

Adam J. Calhoun, Navin Pokala, Ada Tong, James A. J. Fitzpatrick, Tatyana O. Sharpee, and Sreekanth H. Chalasani. Neural mechanisms for evaluating environmental variability in Caenorhabditis elegans. *Neuron*, 86(2):428–441, 2015.

James W. Cannon, William J. Floyd, Richard Kenyon, and Walter R. Parry. Hyperbolic geometry. In *Flavors of Geometry*, volume 31, pp. 59–115. Cambridge University Press, 1997.

Matteo Carandini and Anne K. Churchland. Probing perceptual decisions in rodents. *Nature Neuroscience*, 16(7):824–831, 2013.

Benjamin Paul Chamberlain, James Clough, and Marc Peter Deisenroth. Neural embeddings of graphs in hyperbolic space. *arXiv preprint arXiv:1705.10359*, 2017.

Martina Dal Bello, Alfonso Pérez-Escudero, Frank C. Schroeder, and Jeff Gore. Inversion of pheromone preference optimizes foraging in C. elegans. *eLife*, 10:e58144, 2021.

Justin Fu, Katie Luo, and Sergey Levine. Learning robust rewards with adversarial inverse reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2018.

Octavian Ganea, Gary Bécigneul, and Thomas Hofmann. Hyperbolic neural networks. In *Advances in Neural Information Processing Systems 31 (NeurIPS)*, 2018.

S. Gepshtein, T. D. Albright, E. Leonardis, T. Pereira, P. Jurica, and V. Behner. HUMEX - Human Movement in an Exhibition. `https://doi.org/10.17605/OSF.IO/7F5DU`, October 2026.

Jingyang Ke, Feiyang Wu, Jiyi Wang, Jeffrey Markowitz, and Anqi Wu. Inverse reinforcement learning with switching rewards and history dependency for characterizing animal behaviors. *arXiv preprint arXiv:2501.12633*, 2025.

Dmitri Krioukov, Fragkiskos Papadopoulos, Maksim Kitsak, Amin Vahdat, and Marián Boguñá. Hyperbolic geometry of complex networks. *Physical Review E*, 82:036106, 2010.

Minhae Kwon, Saurabh Daptardar, Paul R. Schrater, and Xaq Pitkow. Inverse rational control with partially observable continuous nonlinear dynamics. In *Advances in Neural Information Processing Systems 33 (NeurIPS)*, pp. 7898–7909, 2020.

Sergey Levine, Zoran Popovic, and Vladlen Koltun. Nonlinear inverse reinforcement learning with gaussian processes. In *Advances in Neural Information Processing Systems 24 (NIPS)*, 2011.

Andriy Mnih and Russ R. Salakhutdinov. Probabilistic matrix factorization. In *Advances in Neural Information Processing Systems 20 (NeurIPS)*, 2007.

Dean Mobbs, Pete C. Trimmer, Daniel T. Blumstein, and Peter Dayan. Foraging for foundations in decision neuroscience: Insights from ethology. *Nature Reviews Neuroscience*, 19(7):419–427, 2018.

Andrew Y. Ng and Stuart J. Russell. Algorithms for inverse reinforcement learning. In *Proceedings of the Seventeenth International Conference on Machine Learning (ICML)*, pp. 663–670, 2000.

Maximilian Nickel and Douwe Kiela. Poincaré embeddings for learning hierarchical representations. In *Advances in Neural Information Processing Systems 30 (NeurIPS)*, 2017.

Maximilian Nickel and Douwe Kiela. Learning continuous hierarchies in the lorentz model of hyperbolic geometry. *arXiv preprint arXiv:1806.03417*, 2018.

Aleksandar Poleksic. Hyperbolic matrix factorization improves prediction of drug-target associations. *Scientific Reports*, 13:959, 2023. doi: 10.1038/s41598-023-27995-5.

Anoop Praturu and Tatyana O. Sharpee. Adaptive data embedding for curved spaces. *iScience*, 27 (12):111266, 2024. doi: 10.1016/j.isci.2024.111266.

Anoop Praturu and Tatyana O. Sharpee. Distance-based logistic matrix factorization. *Neural Computation*, 37(10):1863–1876, 2025. doi: 10.1162/neco.a.25.

Deepak Ramachandran and Eyal Amir. Bayesian inverse reinforcement learning. In *Proceedings of the 20th International Joint Conference on Artificial Intelligence (IJCAI)*, pp. 2586–2591, 2007.

Nathan D. Ratliff, J. Andrew Bagnell, and Martin A. Zinkevich. Maximum margin planning. In *Proceedings of the 23rd International Conference on Machine Learning (ICML)*, pp. 729–736, 2006.

Gautam Reddy. Reinforcement waves as a mechanism for discontinuous learning. *bioRxiv preprint*, 2022.

Nicholas A. Roy, Ji Hyun Bak, Athena Akrami, Carlos D. Brody, and Jonathan W. Pillow. Extracting the dynamics of behavior in sensory decision-making experiments. *Neuron*, 109(4):597–610, 2021.

Nicole C. Rust, Odelia Schwartz, J. Anthony Movshon, and Eero P. Simoncelli. Spatiotemporal elements of macaque v1 receptive fields. *Neuron*, 46(6):945–956, 2005.

Matthias Schultheis, Dominik Straub, and Constantin A. Rothkopf. Inverse optimal control adapted to the noise characteristics of the human sensorimotor system. *arXiv preprint arXiv:2110.11130*, 2021.

Odelia Schwartz, Jonathan W. Pillow, Nicole C. Rust, and Eero P. Simoncelli. Spike-triggered neural characterization. *Journal of Vision*, 6(4):484–507, 2006.

Tatyana O. Sharpee. Computational identification of receptive fields. *Annual Review of Neuroscience*, 36:103–120, 2013.

David W. Stephens and John R. Krebs. *Foraging Theory*. Monographs in Behavior and Ecology. Princeton University Press, 1986.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems 30 (NeurIPS)*, 2017.

Benjamin Wilson and Matthias Leimeister. Gradient descent in hyperbolic space. *arXiv preprint arXiv:1805.08207*, 2018.

Robert C. Wilson and Anne G. E. Collins. Ten simple rules for the computational modeling of behavioral data. *eLife*, 8:e49547, 2019.

Dennis Wu, Yi-Chun Hung, Braden Yuille, James E. Fitzgerald, and Han Liu. Hyperbolic neural population geometry benefits computation. In *International Conference on Machine Learning (ICML)*, 2026.

Markus Wulfmeier, Peter Ondruska, and Ingmar Posner. Maximum entropy deep inverse reinforcement learning. *arXiv preprint arXiv:1507.04888*, 2015.

K. Xu, Z. Qin, G. Wang, K. Huang, S. Ye, and H. Zhang. Collision-free LSTM for human trajectory prediction. In *International Conference on Multimedia Modeling (MMM)*, pp. 106–116. Springer, Cham, February 2018.

Shoichiro Yamaguchi, Honda Naoki, Muneki Ikeda, Yuki Tsukada, Shunji Nakano, Ikue Mori, and Shin Ishii. Identification of animal behavioral strategies by inverse reinforcement learning. *PLOS Computational Biology*, 14(5):e1006122, 2018.

Huanqiu Zhang, P. Dylan Rich, Albert K. Lee, and Tatyana O. Sharpee. Hippocampal spatial representations exhibit a hyperbolic geometry that expands with experience. *Nature Neuroscience*, 26: 131–139, 2023.

Yuansheng Zhou, Brian H. Smith, and Tatyana O. Sharpee. Hyperbolic geometry of the olfactory space. *Science Advances*, 4(8):eaaq1458, 2018.

Brian D. Ziebart, J. Andrew Bagnell, and Anind K. Dey. Maximum entropy inverse reinforcement learning. In *Proceedings of the 23rd National Conference on Artificial Intelligence (AAAI) - Volume 3*, pp. 1433–1438. AAAI Press, 2008.

A Appendix

A.1 Euclidean Distance

We choose the following reward function $R(s_t) = -\frac{1}{2}d^2$ to use as a metric for the Euclidean distance between the feature vector $\vec{\phi}(s_t)$ and the reward parameters $\vec{\theta}$. The reward function is defined as follows: $R(s_t) = \vec{\phi}(s_t) \cdot \vec{\theta}$. We wish to find the parameters of the reward function that maximize the probability of seeing the demonstrations, $P(\mathcal{D}|\theta)$. We derive the loss function for the Euclidean distance reward model as follows:

$$\mathcal{L}(\theta) = -\frac{1}{N}\log P(\mathcal{D}|\theta) \tag{11}$$

$$= -\frac{1}{N}\log \prod_{i=1}^{N} P(\tau_i|\theta) \tag{12}$$

$$= -\frac{1}{N}\sum_{i=1}^{N} \log P(\tau_i|\theta) \tag{13}$$

$$= -\frac{1}{N}\sum_{i=1}^{N} \log \frac{e^{R(\tau_i)}}{Z(\theta)} \tag{14}$$

$$= -\frac{1}{N}\sum_{i=1}^{N} \log(e^{R(\tau_i)}) + \frac{1}{N}\sum_{i=1}^{N} \log Z(\theta) \tag{15}$$

$$= \frac{1}{N}\sum_{i=1}^{N} \log Z(\theta) - \frac{1}{N}\sum_{i=1}^{N} \log(e^{R(\tau_i)}) \tag{16}$$

$$= \frac{1}{N}\sum_{i=1}^{N} \log Z(\theta) - \frac{1}{N}\sum_{i=1}^{N} R(\tau_i) \tag{17}$$

$$= \frac{1}{N}\sum_{i=1}^{N} \log Z(\theta) - \frac{1}{N}\sum_{i=1}^{N} \left[\sum_{t=1}^{T} \vec{\phi}(s_t) \cdot \vec{\theta} - \frac{1}{2}\left(\|\vec{\phi}\|^2 + \|\vec{\theta}\|^2\right)\right] \tag{18}$$

$$= \log Z(\theta) - \frac{1}{N}\sum_{i=1}^{N}\sum_{t=1}^{T} \vec{\phi}(s_t) \cdot \vec{\theta} - \frac{1}{2}(\|\vec{\phi}\|^2 + \|\vec{\theta}\|^2) \tag{19}$$

$$= \log Z(\theta) - \mathbb{E}_{\mu_E}\left[\vec{\phi}(s_t) \cdot \vec{\theta} - \frac{1}{2}(\|\vec{\phi}\|^2 + \|\vec{\theta}\|^2)\right] \tag{20}$$

$$\mathcal{L}(\theta) = \log Z(\theta) - \mathbb{E}_{\mu_E}\left[\vec{\phi}(s_t) \cdot \vec{\theta} - \frac{1}{2}(\|\vec{\theta}\|^2)\right] \tag{21}$$

We provide the gradient of the loss function with respect to θ:

$$(\nabla_\theta \mathcal{L}(\theta))_i = \frac{\partial}{\partial \theta_i}\log Z(\theta) - \frac{\partial}{\partial \theta_i}\mathbb{E}_{\mu_E}\left[\vec{\phi}(s_t) \cdot \vec{\theta} - \frac{1}{2}(\|\vec{\theta}\|^2)\right] \tag{22}$$

We look at the first term of the gradient:

$$= \frac{\partial}{\partial \theta_i} \log Z(\theta) \tag{23}$$

$$= \frac{1}{Z} \frac{\partial}{\partial \theta_i} \sum_{\tau} e^{R(\tau)} \tag{24}$$

$$= \frac{1}{Z} \sum_{\tau} \frac{\partial e^{R(\tau)}}{\partial R(\tau)} \cdot \frac{\partial R(\tau)}{\partial \theta_i} \tag{25}$$

$$= \frac{1}{Z} \sum_{\tau} e^{R(\tau)} \cdot \frac{\partial R(\tau)}{\partial \theta_i} \tag{26}$$

$$= \frac{1}{Z} \sum_{\tau} e^{R(\tau)} \cdot \frac{\partial}{\partial \theta_i} \sum_{t=1}^{T} \left[\vec{\phi}(s_t) \cdot \vec{\theta} - \frac{1}{2}(\|\vec{\phi}\|^2 + \|\vec{\theta}\|^2) \right] \tag{27}$$

$$= \frac{1}{Z} \sum_{\tau} e^{R(\tau)} \sum_{t=1}^{T} \left[\vec{\phi}(s_t) - \vec{\theta} \right] \tag{28}$$

$$= \mathbb{E}_{\mu_\pi} \left[\sum_{t=1}^{T} \vec{\phi}(s_t) - \vec{\theta} \right] \tag{29}$$

We then take the second term of the gradient in the following way:

$$= \frac{\partial}{\partial \theta_i} - \mathbb{E}_{\mu_E} \left[\vec{\phi}(s_t) \cdot \vec{\theta} - \frac{1}{2}(\|\vec{\theta}\|^2) \right] \tag{30}$$

$$= -\mathbb{E}_{\mu_E} \left[\vec{\phi}(s_t) - \vec{\theta} \right] \tag{31}$$

This gives us the following gradient of the loss function with respect to θ using our Euclidean Distance metric for the reward function:

$$(\nabla_\theta \mathcal{L}(\theta))_i = \mathbb{E}_{\mu_\pi} \left[\vec{\phi}(s_t) - \vec{\theta} \right] - \mathbb{E}_{\mu_E} \left[\vec{\phi}(s_t) - \vec{\theta} \right] \tag{32}$$

A.2 Hyperbolic Distance

The reward metric is still $R(s_t) = -\frac{1}{2}d^2$. We use the Lorentzian model where $\mathbb{H}^d$ is represented as $\mathbb{R}^{d+1}$; we define the Minkowski inner product on $\mathbb{R}^{d+1}$ as $\langle \vec{\theta}, \vec{\phi} \rangle = -\theta_0\phi_0 + \sum_{n=1}^{d} \theta_n\phi_n$, and we define the distance as follows: $d = \operatorname{arcosh}(-\langle \vec{\theta}, \vec{\phi}(s_t) \rangle)$, and therefore $R(s_t) = -\frac{1}{2}[\operatorname{arcosh}(-\langle \vec{\theta}, \vec{\phi}(s_t) \rangle)]^2$.

$$\mathcal{L}(\theta) = -\frac{1}{N}\log P(\mathcal{D}|\theta) \tag{33}$$

$$= -\frac{1}{N}\log \prod_{i=1}^{N} P(\tau_i|\theta) \tag{34}$$

$$= -\frac{1}{N}\sum_{i=1}^{N} \log P(\tau_i|\theta) \tag{35}$$

$$= -\frac{1}{N}\sum_{i=1}^{N} \log \frac{e^{R(\tau_i)}}{Z(\theta)} \tag{36}$$

$$= -\frac{1}{N}\sum_{i=1}^{N} \log(e^{R(\tau_i)}) + \frac{1}{N}\sum_{i=1}^{N} \log Z(\theta) \tag{37}$$

$$= \frac{1}{N}\sum_{i=1}^{N} \log Z(\theta) - \frac{1}{N}\sum_{i=1}^{N} R(\tau_i) \tag{38}$$

$$= \log Z(\theta) - \frac{1}{N}\sum_{i=1}^{N}\sum_{t=1}^{T} -\frac{1}{2}[\text{arcosh}(\theta_0\phi_0 - \sum_{n=1}^{d} \theta_n\phi_n(s_t))]^2 \tag{39}$$

$$\mathcal{L}(\theta) = \log Z(\theta) - \mathbb{E}_{\mu_E} R(\tau_i) \tag{40}$$

The gradient of the loss can be written as follows:

$$\mathcal{L}(\theta) = \frac{\partial}{\partial\theta_i}\log Z(\theta) - \frac{\partial}{\partial\theta_i}\mathbb{E}_{\mu_E} R(\tau_i) \tag{41}$$

$$\tag{42}$$

We address the first term of the gradient below, first with respect to θ_0 then with respect to θ_n for $n = 1, ..., d$:

$$= \frac{\partial}{\partial\theta_i}\log Z(\theta) \tag{43}$$

$$= \frac{1}{Z}\frac{\partial}{\partial\theta_i}\sum_{\tau} e^{R(\tau)} \tag{44}$$

$$= \frac{1}{Z}\frac{\partial}{\partial\theta_i}\sum_{\tau} e^{\sum_{t=1}^{T} d^2} \tag{45}$$

$$= \frac{1}{Z}\frac{\partial}{\partial\theta_i}\sum_{\tau} e^{\sum_{t=1}^{T} -\frac{1}{2}d^2} \tag{46}$$

$$\tag{47}$$

if $x = -(\theta_0\phi_0 - \sum_{n=1}^{d} \theta_n\phi_n(s_t))$ and $d = \text{arcosh}(x)$

$$\tag{48}$$

$$= \frac{1}{Z}\sum_{\tau} \frac{\partial e^{R(\tau)}}{\partial \frac{-1}{2}d^2} \cdot \frac{\partial -\frac{1}{2}d^2}{\partial d} \cdot \frac{\partial d}{\partial x} \cdot \frac{\partial x}{\partial \theta_i} \tag{49}$$

$$\tag{50}$$

We use the Hyperbolic identity $\cosh^2 d - 1 = \sinh^2 d$ and therefore $\sinh(d) = \sqrt{\cosh^2(d) - 1}$

(51)

$$\frac{\partial}{\partial \theta_0} \log Z(\theta) = \frac{1}{Z} \sum_{i=1}^{N} \sum_{t=0}^{T} \frac{-de^{-\frac{1}{2}d^2}\phi_0}{\sinh(d)} \tag{52}$$

$$\frac{\partial}{\partial \theta_n} \log Z(\theta) = \frac{1}{Z} \sum_{i=1}^{N} \sum_{t=0}^{T} \frac{de^{-\frac{1}{2}d^2}\phi_n}{\sinh(d)} \tag{53}$$

The second term of the gradient can be written as follows:

$$= \frac{\partial}{\partial \theta_i} - \mathbb{E}_{\mu_E} R(\tau_i) \tag{54}$$

$$= -\frac{1}{N} \sum_{i=1}^{N} \sum_{t=0}^{T} \frac{\partial}{\partial \theta_i} R(s_t) \tag{55}$$

$$= -\frac{1}{N} \sum_{i=1}^{N} \sum_{t=0}^{T} \frac{\partial - \frac{1}{2}d^2}{\partial d} \cdot \frac{\partial d}{\partial x} \cdot \frac{\partial x}{\partial \theta_i} \tag{56}$$

$$\frac{\partial}{\partial \theta_0} - \mathbb{E}_{\mu_E} R(\tau_i) = -\frac{1}{N} \sum_{i=1}^{N} \sum_{t=0}^{T} \frac{-d\phi_0}{\sinh(d)} \tag{57}$$

$$\frac{\partial}{\partial \theta_n} - \mathbb{E}_{\mu_E} R(\tau_i) = \frac{1}{N} \sum_{i=1}^{N} \sum_{t=0}^{T} \frac{-d\phi_n}{\sinh(d)} \tag{58}$$

(59)

We combine the first and second derivative terms:

$$\partial_{\theta_i} \left(\log Z(\theta) - \mathbb{E}_{\mu_E} R(\tau_i) \right) = \begin{bmatrix} \frac{\partial}{\partial \theta_0} : \frac{1}{Z} \sum_{i=1}^{N} \sum_{t=0}^{T} \frac{-de^{-\frac{1}{2}d^2}\phi_0}{\sinh(d)} - \frac{1}{N} \sum_{i=1}^{N} \sum_{t=0}^{T} \frac{-d\phi_0}{\sinh(d)} \\ \\ \frac{\partial}{\partial \theta_n} : \frac{1}{Z} \sum_{i=1}^{N} \sum_{t=0}^{T} \frac{de^{-\frac{1}{2}d^2}\phi_n}{\sinh(d)} + \frac{1}{N} \sum_{i=1}^{N} \sum_{t=0}^{T} \frac{-d\phi_n}{\sinh(d)} \end{bmatrix}, \tag{60}$$

where $\partial_{\theta_i} \left(\log Z(\theta) - \mathbb{E}_{\mu_E} R(\tau_i) \right)$ denotes the naive (Euclidean-sense) partial derivative of the loss. Simplifying each row of Equation 60 gives

$$\partial_{\theta_i} \left(\log Z(\theta) - \mathbb{E}_{\mu_E} R(\tau_i) \right) = \begin{bmatrix} \frac{\partial}{\partial \theta_0} : -\mathbb{E}_{\mu_\pi} \frac{d\phi_0}{\sinh(d)} + \mathbb{E}_{\mu_E} \frac{d\phi_0}{\sinh(d)} \\ \\ \frac{\partial}{\partial \theta_n} : \mathbb{E}_{\mu_\pi} \frac{d\phi_n}{\sinh(d)} - \mathbb{E}_{\mu_E} \frac{d\phi_n}{\sinh(d)} \end{bmatrix}. \tag{61}$$

Note the θ_0 row is the *negative* of the θ_n row's sign pattern ($-\mu_\pi + \mu_E$ versus $\mu_\pi - \mu_E$): this is a direct consequence of the Minkowski signature, not an inconsistency – $\partial x / \partial \theta_0 = \phi_0$ while $\partial x / \partial \theta_n = -\phi_n$ (Section A.2, above), so the chain rule contributes an extra sign flip for the time-like coordinate that it does not for the space-like ones.

Because the ambient space carries the Minkowski metric $g = \mathrm{diag}(-1, 1, \ldots, 1)$ rather than the Euclidean one, the gradient actually used for Riemannian optimization (Section A.3, Appendix A.3) is not the naive gradient of Equation 61 directly, but its image under $g^{-1} = g$, i.e. the same vector with its time-like (θ_0) component negated. This is a separate step, applied *after* the derivative itself is fully computed – it is the same step Praturu & Sharpee (2025) make explicit when constructing their analogous "unconstrained gradient" $\nabla_u^{d+1} \mathcal{L}_H = (-\partial \mathcal{L}_H / \partial u_0,\ \partial \mathcal{L}_H / \partial u_1,\ \ldots)$ (their Eq. 2.14) only after computing $\partial \mathcal{L}_H / \partial u_0$ by ordinary calculus (their Eq. 2.16), noting that "the minus sign

on the 0th component of the derivative arises from the signature of the Minkowski inner product." Negating the θ_0 row of Equation 61 – i.e. flipping the sign of every term in that row, turning $-\mu_\pi + \mu_E$ into $\mu_\pi - \mu_E$ – gives the ambient gradient used throughout the rest of this paper:

$$\nabla_\theta^{d+1} = \begin{bmatrix} \dfrac{\partial}{\partial \theta_0} : \mathbb{E}_{\mu_\pi} \dfrac{d\phi_0}{\sinh(d)} - \mathbb{E}_{\mu_E} \dfrac{d\phi_0}{\sinh(d)} \\ \dfrac{\partial}{\partial \theta_n} : \mathbb{E}_{\mu_\pi} \dfrac{d\phi_n}{\sinh(d)} - \mathbb{E}_{\mu_E} \dfrac{d\phi_n}{\sinh(d)} \end{bmatrix} \tag{62}$$

(63)

which is the exact gradient for the Hyperbolic reward (Section 2.3). In practice, $\sinh(d)$ grows exponentially and $1/\sinh(d)$ collapses toward zero for states far from θ, so, following Praturu & Sharpee (2025) who report the same difficulty for the analogous term in their model, we drop the $1/\sinh(d)$ factor and retain only the linear d dependence:

$$\nabla_\theta \mathcal{L} \approx \frac{\partial}{\partial \theta_0} : \ \mathbb{E}_{\mu_\pi}[d\,\phi_0] - \mathbb{E}_{\mu_E}[d\,\phi_0] \tag{64}$$

$$\frac{\partial}{\partial \theta_n} : \ \mathbb{E}_{\mu_\pi}[d\,\phi_n] - \mathbb{E}_{\mu_E}[d\,\phi_n]$$

All Hyperbolic results reported in this paper use this practical gradient.

This practical gradient is the starting point for the tangent-space projection of Section A.3, below.

A.3 Riemannian Optimization: Tangent-Space Projection and the Exponential Map

Equation 63 (or its practical approximation, Equation 64) is an *ambient* gradient in $\mathbb{R}^{d+1}$: it does not account for the constraint $\langle \vec{\theta}, \vec{\theta} \rangle_{\mathcal{M}} = -1$ that keeps θ on the hyperboloid. Taking $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$ directly would therefore move θ off the manifold. We instead perform Riemannian gradient descent (Nickel & Kiela, 2018; Wilson & Leimeister, 2018; Praturu & Sharpee, 2025): project the ambient gradient onto the tangent space at θ, then take a step along the geodesic in that direction via the exponential map, so that every update remains exactly on $\mathbb{H}^d$.

Tangent-space projection. The tangent space at θ is $T_\theta \mathbb{H}^d = \{v \in \mathbb{R}^{d+1} : \langle \vec{\theta}, v \rangle_{\mathcal{M}} = 0\}$. The Riemannian gradient is obtained from the ambient gradient by adding back its component along θ, using the Minkowski inner product in place of the Euclidean one:

$$\nabla_\theta^{\mathcal{H}} \mathcal{L} = \nabla_\theta \mathcal{L} + \left\langle \vec{\theta}, \nabla_\theta \mathcal{L} \right\rangle_{\mathcal{M}} \vec{\theta}. \tag{65}$$

Exponential map. For a tangent vector $y \in T_\theta \mathbb{H}^d$, the exponential map

$$\mathrm{Exp}_\theta(y) = \cosh\!\left(\sqrt{\langle y, y \rangle_{\mathcal{M}}}\right) \vec{\theta} + \sinh\!\left(\sqrt{\langle y, y \rangle_{\mathcal{M}}}\right) \frac{y}{\sqrt{\langle y, y \rangle_{\mathcal{M}}}} \tag{66}$$

takes a step of geodesic length $\sqrt{\langle y, y \rangle_{\mathcal{M}}}$ along $\mathbb{H}^d$ in the direction of y. We replace the Euclidean update rule with

$$\theta \leftarrow \mathrm{Exp}_\theta\left(-\eta\, \nabla_\theta^{\mathcal{H}} \mathcal{L}\right), \tag{67}$$

for learning rate η, i.e. $y = -\eta \nabla_\theta^{\mathcal{H}} \mathcal{L}$ in Equation 66.

For the radius-R hyperboloid (Section 2.4), the Riemannian gradient correction (Equation 65) picks up a compensating factor of $1/R^2$,

$$\nabla_\theta^{\mathcal{H}} \mathcal{L} = \nabla_\theta \mathcal{L} + \frac{1}{R^2} \left\langle \vec{\theta}, \nabla_\theta \mathcal{L} \right\rangle_{\mathcal{M}} \vec{\theta}, \tag{68}$$

since $\langle \vec{\theta}, \vec{\theta} \rangle_{\mathcal{M}} = -R^2$ rather than -1, and the exponential map (Equation 66) becomes

$$\mathrm{Exp}_\theta(y) = \cosh\!\left(\frac{\|y\|_{\mathcal{M}}}{R}\right) \vec{\theta} + R \sinh\!\left(\frac{\|y\|_{\mathcal{M}}}{R}\right) \frac{y}{\|y\|_{\mathcal{M}}}, \qquad \|y\|_{\mathcal{M}} \equiv \sqrt{\langle y, y \rangle_{\mathcal{M}}}, \tag{69}$$

where $R = 1$ recovers Equations 65–66.

A.4 Numerical Stability of the Exponential-Map Step

The exponential-map update (Equation 67) is only numerically safe for moderate step sizes. Two failure modes appear in practice, both traceable to the same cause: as θ drifts away from the origin of the hyperboloid over the course of training, curvature makes a fixed step size η eventually produce an unstable update, regardless of how θ got there. (We verified that this instability is intrinsic to the fixed step size rather than a sampling-noise artifact from the policy rollouts used to estimate μ_π: increasing the number of rollouts by an order of magnitude did not change when the instability appeared.) Writing $y = -\eta \nabla^{\mathcal{H}}_\theta \mathcal{L}$ for the proposed tangent step before it is applied:

1. **Loss of precision in $\langle y, y \rangle_{\mathcal{M}}$.** This Minkowski norm is computed as a difference of two large, nearly equal floating-point sums once y's components exceed roughly 10^8, so it loses precision to catastrophic cancellation and can come out negative – i.e. not a valid tangent vector – even when y is mathematically valid.
2. **Overflow in $\cosh$ / $\sinh$.** Even when $\langle y, y \rangle_{\mathcal{M}}$ is computed correctly, $\cosh(\sqrt{\langle y, y \rangle_{\mathcal{M}}})$ and $\sinh(\sqrt{\langle y, y \rangle_{\mathcal{M}}})$ overflow double-precision floating point once $\sqrt{\langle y, y \rangle_{\mathcal{M}}}$ is roughly 709 or larger, producing `inf`/`NaN` values that propagate into the softmax policy computed on the next iteration.

We address both failure modes at their common source by capping the geodesic step length before either $\langle y, y \rangle_{\mathcal{M}}$ or the exponential map is evaluated: given a maximum step norm $S_{\max}$, if $\eta \left\| \nabla^{\mathcal{H}}_\theta \mathcal{L} \right\|_2 > S_{\max}$ (using the ordinary Euclidean norm, which is cheap to compute and tracks the eventual geodesic step length monotonically), we rescale

$$\eta \leftarrow \eta \cdot \frac{S_{\max}}{\eta \left\| \nabla^{\mathcal{H}}_\theta \mathcal{L} \right\|_2} \tag{70}$$

before taking the step – i.e. the learning rate is only ever reduced, never increased, and the rescaling happens before any Hyperbolic quantity is computed. We use $S_{\max} = 5$ geodesic units throughout.

A.5 Mixed Second Difference

To summarize non-additivity in the reward surface as a single number per feature pair (Section 3.3), we compute the mixed second difference of the reward surface, evaluated at two corner values $\text{hi} = 0.9$ and $\text{lo} = 0.1$ of each feature:

$$\Delta^2 R = R(\text{hi}, \text{hi}) - R(\text{hi}, \text{lo}) - R(\text{lo}, \text{hi}) + R(\text{lo}, \text{lo}). \tag{71}$$

This is the discrete analogue of the mixed partial derivative $\partial^2 R / \partial \phi_i \partial \phi_j$, and like the residual of the two-way ANOVA-style fit (Section 3.3), it is exactly zero whenever R is additively separable in ϕ_i and ϕ_j. Throughout the rest of the paper we refer to this quantity as the conjunctive effect.

A.6 Museum Exhibit Reward Magnitude

Figure 8 shows the museum floorplan with each exhibit colored by $|\theta_k|$, the magnitude of that exhibit's own fitted weight (Hyperbolic, κ=-4), i.e. how much reward that exhibit contributes on its own, uncoupled from any other exhibit.

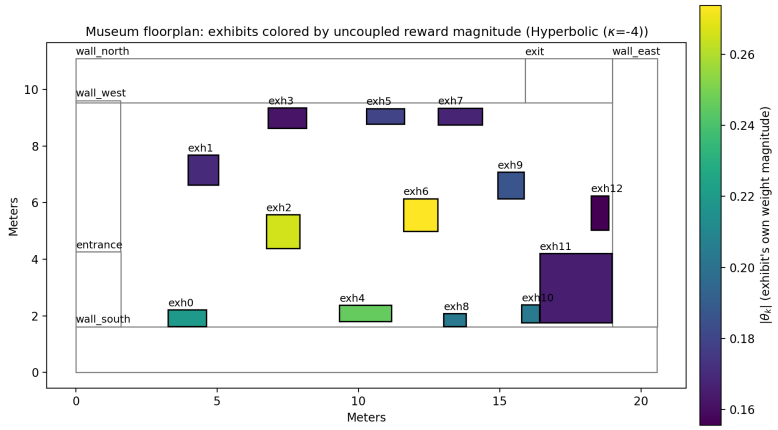


Figure 8: Museum floorplan with each exhibit colored by its own (uncoupled) fitted reward weight magnitude $|\theta_k|$, Hyperbolic (κ=-4).

A.7 Permutation (Mantel) Test for Reward-Coupling Alignment

The exhibit-pair alignment test in Section 3.3 correlates $\mathrm{co_attendance}(i, j)$, the real behavioral correlation between how closely visitors approach exhibits i and j in the same trajectory, against each geometry's fitted conjunctive effect for that pair. Because the 78 exhibit pairs are built from only 13 exhibits (each exhibit appears in 12 of the 78 pairs), the pairs are not independent samples, so the naive p-values from that correlation are likely optimistic. We correct for this with a Mantel-style permutation test: we permute the 13 exhibit labels jointly across rows and columns of the $\mathrm{co_attendance}$ matrix (holding the conjunctive-effect matrix fixed), recompute the correlation, and repeat for 10,000 shuffles to build an empirical null distribution. This preserves the pairwise dependency structure among the 78 correlations (since permuting exhibit labels permutes all 12 pairs involving a given exhibit together) and is the standard way to test correlations computed over pairs drawn from a smaller set of underlying units.

The resulting correlations and permutation-corrected p-values are reported in Table 1 below, corresponding to the scatter plots in Figure 7 in the main text (Section 3.3). As a further check, we also residualize both $\mathrm{co_attendance}$ and the fitted conjunctive effect against physical exhibit-to-exhibit distance before correlating (partial correlation columns), to rule out spatial proximity alone as a confound; this leaves the pattern essentially unchanged for every geometry (e.g., for Hyperbolic (κ=-4), partial $r = +0.539$ vs. raw $r = +0.516$), with the same three curvatures remaining significant under permutation correction. We also report the naive (non-permutation) p-value alongside the corrected one to make explicit how much the non-independence of the 78 pairs inflates apparent significance: e.g., Hyperbolic (κ=-4)'s naive $p \approx 1.3 \times 10^{-6}$ becomes $p = 0.0026$ once corrected, still significant but three orders of magnitude less extreme.

Table 1: Alignment between real exhibit co-attendance and each geometry's fitted conjunctive effect: raw correlation with its naive (non-permutation) and permutation-corrected p-values (10,000 shuffles of exhibit labels), and the same three quantities after residualizing both variables against physical exhibit-to-exhibit distance (partial correlation).

	Raw			Partial (distance-controlled)		
Geometry	r	naive p	perm. p	r	naive p	perm. p
Classic	-0.070	0.54	0.48	-0.048	0.68	0.65
Euclidean	+0.075	0.51	0.53	+0.058	0.62	0.63
Hyperbolic (κ=-25)	+0.477	1.0×10^{-5}	0.0021	+0.507	2.2×10^{-6}	0.0011
Hyperbolic (κ=-4)	+0.516	1.3×10^{-6}	0.0026	+0.539	3.5×10^{-7}	0.0019
Hyperbolic (κ=-1)	+0.419	1.3×10^{-4}	0.0034	+0.421	1.2×10^{-4}	0.0035
DeepIRL	+0.205	0.071	0.084	+0.228	0.045	0.056

This permutation test only corrects for the non-independence among the 78 *exhibit-pair* samples; it does not address a separate form of non-independence at the *trajectory* level (each visitor trajectory contributes to the $\mathrm{co_attendance}$ value of multiple exhibit pairs), which is not addressed here.